E-learning Purchase Database

1. Project Objective

The main objective of this project is to analyze **E-learning platform purchase data using MySQL** and derive meaningful business insights from learner, course, and purchase information.

An online learning platform offers various digital courses to learners across different countries.

Learners can purchase multiple courses, and each course belongs to a specific category.

The management team wants to understand **sales trends, learner behavior, and popular course categories** from the available purchase data.

To address this requirement, a relational database is designed using MySQL with three main tables: **learners, courses, and purchases**. SQL queries are then used to combine the data, calculate revenue and spending, identify purchasing patterns, and generate useful insights for decision-making.

2. Problem Statement

- Analyze learner purchasing behavior and spending patterns.
- Identify the most purchased courses.
- Analyze category-wise revenue and learner participation.
- Identify learners who purchase courses across multiple categories.
- Identify courses that have never been purchased.
- Apply SQL concepts such as Joins, Aggregations, Subqueries, CTEs, CASE expressions, NULL handling, and Views.

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3. Attribute (Column /Features) Details:

Table 1: learners

Attribute Name	Data Type	Description
learner_id	Int (Primary Key)	Unique identifier assigned to each learners
full_name	Varchar (100)	Name of the learners
Country	Varchar (50)	Name of the country

Table 2: courses

Attribute Name	Data Type	Description
course_id	Int (Primary Key)	Unique identifier assigned to each course
course_name	Varchar (200)	Name of the course
category	Varchar (200)	Name of the course category
unit_price	Decimal (10,2)	Price of the course

Table 3: purchases

Attribute Name	Data Type	Description
purchase_id	Int (Primary Key) (Auto Incremented)	Unique identifier assigned to each learners purchaseid
Learner_id	Int (Foreign Key)	Unique identifier assigned to each learners referenced to ForeignKey Constrint
course_id	Int (Foreign Key)	Unique identifier assigned to each courses referenced to ForeignKey Constrint
quantity	Int	Quantity of purchases
purchase-date	Date	Date of purchases of learners

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4. Tools & Technologies

- **MySQL:** DDL, DML Commands, My SQL Constraints, Joins, Aggregate Functions, Subqueries, CTEs, CASE Expression, NULL Handling, Views, GROUP BY & HAVING, and ORDER BY & LIMIT

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5. Data Pre-Processing (MySQL)

Tasks Performed:

1. **Database and Table Creation (CREATE):**

learners table creation

learner_id	full_name	country
1	Anitha Raj	India
2	Priya Charan	India
3	Divya Priya	India
4	Vijay Anand	India
5	Sanjay Kumar	India
6	Swetha Rani	USA
7	Manoj Kumar	USA
8	Deepa Lakshmi	USA
9	Michael Brown	UK
10	Emma Johnson	UK
11	Oliver Jones	Canada
12	Sophia Wilson	Canada
NULL	NULL	NULL

learners 1 courses 2 purchases 3

courses table creation

course_id	course_name	category	unit_price
101	Excel for Beginners	Beginner	2500.00
102	Python Basics	Beginner	3000.00
103	SQL for Beginners	Beginner	2800.00
104	HTML and CSS	Web Development	6000.00
105	JavaScript Fundamentals	Web Development	7000.00
106	React JS Development	Web Development	8000.00
107	Java Programming	Programming	6500.00
108	C,C++ Programming	Programming	7500.00
109	AWS Cloud Fundamentals	Cloud Computing	8000.00
110	Microsoft Azure Basics	Cloud Computing	8000.00
111	Network Security	Cybersecurity	7500.00
112	Cybersecurity Fundamen...	Cybersecurity	8000.00
113	Digital Marketing Basics	Digital Marketing	4000.00
114	Social Media Marketing	Digital Marketing	6000.00
115	UI/UX Design Fundament...	UI/UX Design	3500.00
116	Graphic Design Basics	UI/UX Design	4000.00
NULL	NULL	NULL	NULL

learners 1 courses 2 purchases 3

purchases table creation

purchase_id	learner_id	course_id	quantity	purchase_date
1001	1	101	1	2026-01-05
1002	1	105	2	2026-01-12
1003	1	107	1	2026-02-10
1004	2	102	1	2026-01-08
1005	2	106	1	2026-02-15
1006	2	112	1	2026-03-05
1007	3	103	2	2026-01-15
1008	3	104	1	2026-02-20
1009	4	105	1	2026-01-20
1010	4	110	1	2026-03-01
1011	4	113	1	2026-03-15
1012	5	101	2	2026-01-25
1013	5	108	1	2026-02-25
1014	6	106	1	2026-01-10
1015	6	111	1	2026-02-10
1016	7	105	1	2026-01-18
1017	7	110	2	2026-02-18
1018	8	102	1	2026-01-22
1019	8	109	1	2026-03-10
1020	9	107	1	2026-01-12
1021	9	113	1	2026-02-12
1022	10	103	1	2026-01-28

learners 1 courses 2 purchases 3

2. Data Exploration using joins

- **INNER JOIN--Displays only purchases where the learners and courses exist**

full_name	course_name	category	Total_Amount	purchase_date
Manoj Kumar	Microsoft Azure Basics	Cloud Computing	16000.00	2026-02-18
Anitha Raj	JavaScript Fundamentals	Web Development	14000.00	2026-01-12
Anitha Raj	React JS Development	Web Development	8000.00	2026-04-01
Priya Charan	React JS Development	Web Development	8000.00	2026-02-15
Priya Charan	Cybersecurity Fundamentals	Cybersecurity	8000.00	2026-03-05
Vijay Anand	Microsoft Azure Basics	Cloud Computing	8000.00	2026-03-01
Swetha Rani	React JS Development	Web Development	8000.00	2026-01-10
Manoj Kumar	React JS Development	Web Development	8000.00	2026-04-10
Deepa Lakshmi	AWS Cloud Fundamentals	Cloud Computing	8000.00	2026-03-10
Emma Johnson	Cybersecurity Fundamentals	Cybersecurity	8000.00	2026-02-28
Oliver Jones	Microsoft Azure Basics	Cloud Computing	8000.00	2026-03-20
Vijay Anand	Network Security	Cybersecurity	7500.00	2026-04-05
Sanjay Kumar	C,C++ Programming	Programming	7500.00	2026-02-25
Swetha Rani	Network Security	Cybersecurity	7500.00	2026-02-10
Oliver Jones	C,C++ Programming	Programming	7500.00	2026-01-30
Vijay Anand	JavaScript Fundamentals	Web Development	7000.00	2026-01-20
Manoj Kumar	JavaScript Fundamentals	Web Development	7000.00	2026-01-18
Anitha Raj	Java Programming	Programming	6500.00	2026-02-10
Michael Brown	Java Programming	Programming	6500.00	2026-01-12
Divya Priya	HTML and CSS	Web Development	6000.00	2026-02-20
Sophia Wilson	HTML and CSS	Web Development	6000.00	2026-03-05
Divya Priya	SQL for Beginners	Beginner	5600.00	2026-01-15

3. Core Analytical Queries:

- **Find the top 3 most purchased courses by quantity**

course_id	course_name	Total_Quantity
101	Excel for Beginners	4
105	JavaScript Fundamentals	4
106	React JS Development	4

- **Identify courses never purchased**

learner_id	full_name	course_name	category
NULL	NULL	Social Media Marketing	Digital Marketing
NULL	NULL	UI/UX Design Fundamentals	UI/UX Design
NULL	NULL	Graphic Design Basics	UI/UX Design

4. SubQueries & Correlated Subqueries

- **Display courses whose price is higher than any course in the 'Beginner' category**

course_id	course_name	category	unit_price
104	HTML and CSS	Web Development	6000.00
105	JavaScript Fundamentals	Web Development	7000.00
106	React JS Development	Web Development	8000.00
107	Java Programming	Programming	6500.00
108	C,C++ Programming	Programming	7500.00
109	AWS Cloud Fundamentals	Cloud Computing	8000.00
110	Microsoft Azure Basics	Cloud Computing	8000.00
111	Network Security	Cybersecurity	7500.00
112	Cybersecurity Fundamentals	Cybersecurity	8000.00
113	Digital Marketing Basics	Digital Marketing	4000.00
114	Social Media Marketing	Digital Marketing	6000.00
115	UI/UX Design Fundamentals	UI/UX Design	3500.00
116	Graphic Design Basics	UI/UX Design	4000.00
NULL	NULL	NULL	NULL

- Find learners who spent more than the average spending in their country

learner_id	full_name	country	Total_Spending
1	Anitha Raj	India	31000.00
4	Vijay Anand	India	26500.00
7	Manoj Kumar	USA	31000.00
10	Emma Johnson	UK	10800.00
11	Oliver Jones	Canada	15500.00

5. CTE, CASE, VIEW, and NULL handling:

- **Use a CTE to calculate total spending per learner, then: Display learners with spending above 10,000**

full_name	Total_spending
Anitha Raj	31000.00
Priya Charan	19000.00
Divya Priya	11600.00
Vijay Anand	26500.00
Sanjay Kumar	12500.00
Swetha Rani	15500.00
Manoj Kumar	31000.00
Deepa Lakshmi	11000.00
Michael Brown	10500.00
Emma Johnson	10800.00
Oliver Jones	15500.00

- **NULL Handling ● Display all courses and replace NULL purchase counts with 0 using: IFNULL()**

course_id	course_name	category	Purchase_count
101	Excel for Beginners	Beginner	3
102	Python Basics	Beginner	2
103	SQL for Beginners	Beginner	2
104	HTML and CSS	Web Development	2
105	JavaScript Fundamentals	Web Development	3
106	React JS Development	Web Development	4
107	Java Programming	Programming	2
108	C,C++ Programming	Programming	2
109	AWS Cloud Fundamentals	Cloud Computing	1
110	Microsoft Azure Basics	Cloud Computing	3
111	Network Security	Cybersecurity	2
112	Cybersecurity Fundamen...	Cybersecurity	2
113	Digital Marketing Basics	Digital Marketing	2
114	Social Media Marketing	Digital Marketing	0
115	UI/UX Design Fundament...	UI/UX Design	0
116	Graphic Design Basics	UI/UX Design	0

- **View ● Create a view: category_performance_view**

category	Total_revenue	Number_of_purchases	Average_revenue_per_purchase
Beginner	24400.00	7	3485.71
Web Development	72000.00	9	8000.00
Programming	28000.00	4	7000.00
Cloud Computing	40000.00	4	10000.00
Cybersecurity	31000.00	4	7750.00
Digital Marketing	8000.00	2	4000.00
UI/UX Design	0.00	0	0.00

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6. Key Insights:

The analysis generated ₹2,03,400 in total revenue, with Web Development emerging as the highest-performing category at ₹72,000. Anitha Raj and Manoj Kumar were the highest-spending learners at ₹31,000 each. Excel for Beginners, JavaScript Fundamentals, React JS Development, and Microsoft Azure Basics recorded the highest purchase quantity. These courses have strong current demand and may continue to attract purchases. India has the highest average learner spending. UK has the lowest average learner spending. The analysis also identified UI/UX Design as an area requiring attention due to zero purchases. Based on these findings, targeted marketing, cross-selling, and personalized offers can be used to improve future revenue and learner engagement. Prioritize marketing and course development based on category demand and revenue potential.

7. Conclusions

This project demonstrates how **MySQL can be used to transform E-learning purchase data into meaningful business insights**.

Through joins, aggregations, filtering, subqueries, CTEs, CASE expressions, NULL handling, and views, the analysis provides insights into **learner spending, course popularity, category performance, and purchasing behavior**.

The project also helped strengthen practical SQL skills in **database design, data analysis, query optimization, and result interpretation**. The insights can support the management team in making better decisions, such as focusing marketing efforts on **top-performing course categories** and understanding high-value learners.